FIITJEE

ALL INDIA TEST SERIES

FULL TEST –X

JEE (Advanced)-2020

PAPER – 2

TEST DATE: 20-09-2020

Time Allotted: 3 Hours **Maximum Marks: 186**

General Instructions:

- The test consists of total 54 questions.
- Each subject (PCM) has 18 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Two Sections**: **Section-A & Section-C.**

1. ***Section-A (01 – 08, 19 – 26, 37 – 44)*** *contains 24 multiple choice questions which have* ***one or more than one correct*** *answer. Each question carries* ***+4 marks*** *for all correct answer.*
   *Full Marks : **+4** If only (all) the four option(s) is (are) chosen.*
   *Partial Marks : **+3** If all the four options are correct but ONLY three options are chosen.*
   *Partial Marks : **+2** If three or more options are correct but ONLY two options are chosen and both of which are correct options.*
   *Partial Marks : **+1** If two or more options are correct but ONLY one option is chosen and it is a correct option.*
   *Zero Marks : **0** If none of the options is chosen (i.e. the question is unanswered).*
   *Negative Marks : **–1** In all other cases.*

   Section-A (09 – 12, 27 – 30, 45 – 48) *contains 6 List-Match sets with 12 questions (each set has 2 questions). Each question has 4 statements in LIST-I & 4, 5 or 6 statements in LIST-II. The codes for Lists have choices (A), (B), (C), (D) out of which only one is correct.*
   Each question has ***only one correct*** *answer and carries* ***+3 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

2. ***Section-C (13 – 18, 31 – 36, 49 – 54)*** *contains 18 Numerical answer type questions with answer XXXXX.XX and each question carries* ***+3 marks*** *for correct answer. There is no negative marking.*

Physics

PART – I

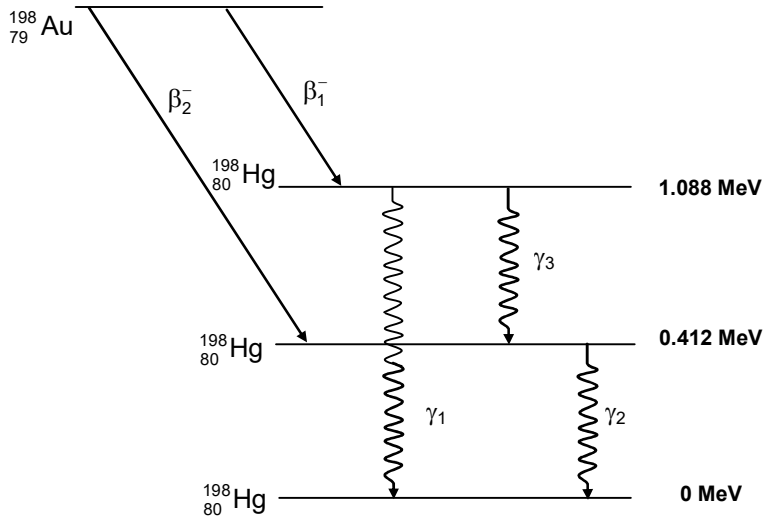
SECTION – A
(One or More than one correct type)

This section contains **08** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

1. Two identical charged particles having charge 10 μC and mass 100 gm each are released on a rough horizontal plane surface at a distance of 1 m from each other. Coefficient of friction between the particles and horizontal plane surface is 0.1 and acceleration due to gravity is 10 m/s^2.
   (A) Maximum speed acquired by each particle is 2 m/s.
   (B) Maximum speed acquired by each particle is $\sqrt{2}$ m/s.
   (C) Magnitude of the displacement of each particle with respect to their initial position when they have their maximum velocity is 1m.
   (D) Initial electrostatic potential energy is totally lost in the form of heat in doing work against the friction force.

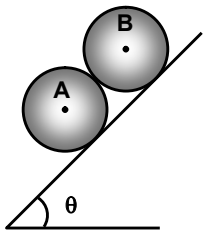
2. Ratio of the radii of nuclei of two hydrogen like atoms is 3 : 2. Assume their nuclei to be having equal no of protons and neutrons. Identify the correct option(s)
   (A) Ratio of orbital radius for K-shell electrons is $\frac{8}{27}$.
   (B) Ratio of velocity of the electrons in K shells is $\frac{27}{8}$.
   (C) Ratio of ionisation energy of the electrons from these two H like atoms is $\frac{729}{64}$
   (D) Ratio of the angular momentum of the electrons orbiting in the first excited state is 1 : 1.

3. In the decay process of a nucleus given below, identify the correct statement.
   [given: m(Au^{198}) = 197.968233 u, m(Hg^{198}) = 197.966760 u] (1u = 931.5 MeV/C^2)



   (A) maximum K.E. of $\beta_1^- = 0.284$ MeV
   (B) maximum K.E. of $\beta_2^- = 0.960$ MeV
   (C) Energy associated with γ_1 is 0.412 MeV
   (D) Energy associated with γ_3 is 0.676 MeV

4. Two solid cylinders A and B are placed in contact on an inclined plane as shown in the diagram. Their dimensions are equal and they are in equilibrium.



Let M_A and M_B be the mass of cylinders A and B respectively.
f_A be friction force between cylinder A and inclined plane
f_B be friction force between cylinder B and inclined plane.
f_{AB} be friction force between cylinder A and cylinder B.
N_{AB} be the normal force between cylinders A and B.
Choose the correct option(s).

(A) $f_A = f_B = f_{AB}$

(B) $N_{AB} = \left(\frac{M_B - M_A}{2}\right) g\sin\theta$

(C) $M_B > M_A$

(D) $f_A > f_B > f_{AB}$

5. The pressure of an ideal gas is written by $P = K\frac{E}{V}$, where E is some type of kinetic energy of random motion, K is a constant and V is the volume of gas. Consider the following statements and mark the correct option(s).

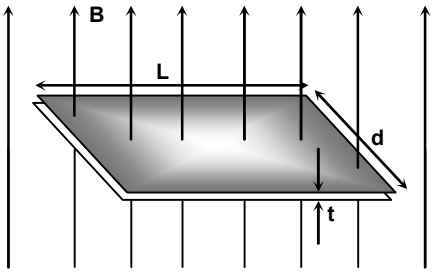
(A) The gas is di-atomic, $K = \frac{2}{5}$ and E = total kinetic energy

(B) The gas is di-atomic, K = 1 and E = rotational kinetic energy

(C) The gas is monoatomic, $K = \frac{2}{3}$ and E = total kinetic energy

(D) The gas is mono atomic or diatomic, $K = \frac{2}{3}$ and E = translational kinetic energy.

6. A long metal strip of thickness t = 1 mm and width d = 1 cm carries current along its length (L). The current in the metal strip is flowing towards the left. It is placed in a uniform transverse magnetic field B = 2 T (as shown in the figure). The potential difference recorded across the width d of the plate is found to be 3×10^{-6} V. Now mark the correct option(s).



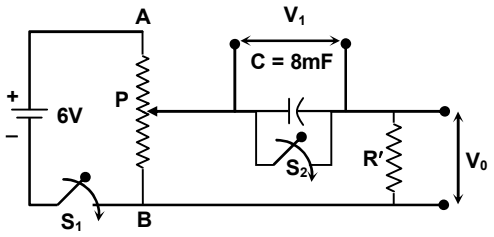
(A) Electric field resulted across the width 'd' is 3×10^{-3} V/m.

(B) Electric field resulted across the width 'd' is 3×10^{-4} V/m.

(C) Magnitude of electric force on the electrons due to the electric field that is resulted due to motion of the electrons inside metal strip is 4.8×10^{-22} N.

(D) The drift velocity of the electron is 1.5×10^{-4} m/s.

7. In the adjacent circuit, AB is a rheostat of 2 kΩ and it is tapped at the central point P. Initially both the switches S_1 and S_2 are closed and potential difference recorded across R′ = 1kΩ is found to be V_0. At t = 0, S_2 is opened and the final potential difference recorded across the 8 mF capacitor is found to be V_1. Then choose the correct option(s).



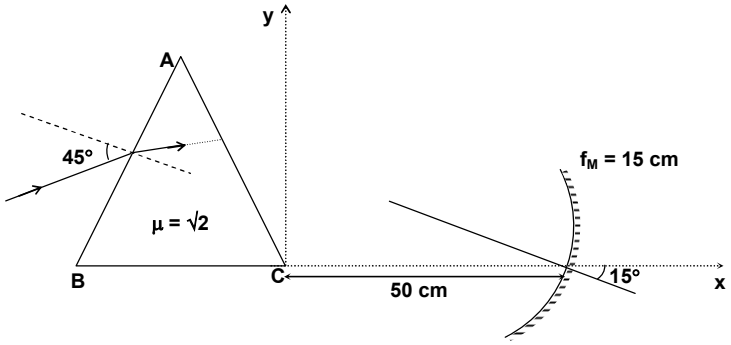
(A) $\frac{V_1}{V_0} = \frac{3}{2}$

(B) $\frac{V_1}{V_0}$ is independent of position of the tapping point P.

(C) maximum rate at which energy is increasing in the capacitor after opening S_2 is 1.5 mW.

(D) the total heat dissipated across R′ after opening S_2 is 24 mJ.

8. A narrow parallel beam of light coming from a distant object is incident on an equilateral triangular prism of refractive index $\mu = \sqrt{2}$ with its base BC along the x-axis. A concave mirror of radius of curvature 30 cm is placed at a distance of 50 cm from the point C of the triangular prism. The principal axis of the mirror is inclined at angle of 15° with x-axis as shown in the diagram. Mark the correct option(s). Assume paraxial object is valid.



(A) Deviation in the beam of light after emerging from the prism is 30°.

(B) Final image formed after reflection from mirror is real, inverted and diminished.

(C) Y-coordinate of the final image formed after reflection from mirror is $\frac{15\left(\sqrt{3}-1\right)}{2\sqrt{2}}$ cm.

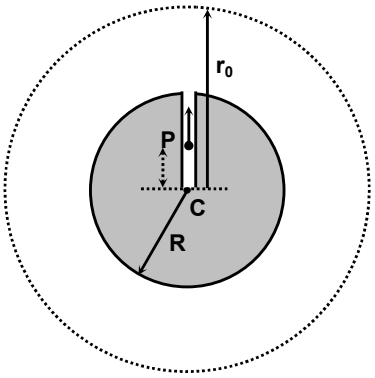
(D) Ray undergoes minimum deviation after passing through the prism.

(Matching List Type)

This section contains **TWO (02)** List-Match Sets. Each List-Match set has **TWO(02)** Multiple Choice Questions. Each List-Match set has two lists: **List-I** and **List-II**. **List-I** has **Four** entries (I), (II), (III) and (IV) and **List-II** has **Six** entries (P), (Q), (R), (S), (T) and (U). **FOUR** options are given in each Multiple Choice Question based On **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.

Answer the following by appropriately matching the list based on the information given in the paragraph.

A satellite is to be launched in a circular orbit of radius r_0 around a planet of radius R. For this two stage rocket is to be used. First stage rocket is used to lift the satellite vertically up to the desired height r_0 (measured from the center C of the planet), after launching it from a launch point P. Distance of launch point P from centre of the planet is r_L. Launch point P lies inside the tunnel as shown in the figure. Second stage rocket is used to impart it a tangential velocity so as to put it in a circular orbit. Let E_1 and E_2 are the energies delivered by the first and second stage of the rocket. Assume that the mass of the rocket is negligible with respect to the mass of the planet and atmospheric resistance to be zero. List-I gives 4 values of r_0 and r_L and List-II gives magnitude of some quantity.



	List – I		List – II
(I)	$r_0 = R$ $r_L = \frac{R}{\sqrt{2}}$	(P)	$\frac{1}{2}$
(II)	$r_0 = \frac{3R}{2}$ $r_L = R$	(Q)	1
(III)	$r_0 = 4R$ $r_L = R$	(R)	$\frac{11}{6}$
(IV)	$r_0 = 4R$ $r_L = \frac{R}{\sqrt{2}}$	(S)	4
		(T)	6
		(U)	8

9. The correct match for the time period of the satellite for the values of r_0 given in List-I in terms of T_0 is

(Take $T_0 = \frac{2\pi}{\sqrt{GM}}R^{3/2}$ and $\sqrt{6} = \frac{22}{9}$ if required)

Options

(A) $I \to Q$, $II \to R$, $III \to P$, $IV \to U$
(B) $I \to Q$, $II \to P$, $III \to U$, $IV \to U$
(C) $I \to Q$, $II \to T$, $III \to U$, $IV \to P$
(D) $I \to Q$, $II \to R$, $III \to U$, $IV \to U$

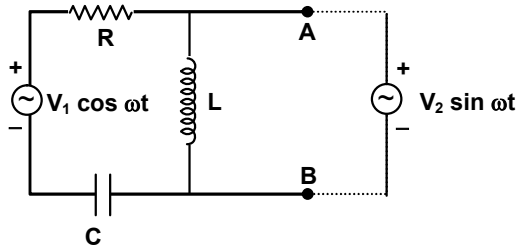
10. The correct match for $\frac{E_1}{E_2}$ for the values of r_0 and r_L given in List-I is

Options

(A) $I \rightarrow S, II \rightarrow T, III \rightarrow U, IV \rightarrow P$
(B) $I \rightarrow P, II \rightarrow Q, III \rightarrow T, IV \rightarrow U$
(C) $I \rightarrow S, II \rightarrow T, III \rightarrow R, IV \rightarrow P$
(D) $I \rightarrow P, II \rightarrow S, III \rightarrow Q, IV \rightarrow T$

Answer the following by appropriately matching the list based on the information given in the paragraph.

A series LCR circuit is fed by an alternating voltage $v_1 \cos \omega t$. Another alternating voltage source $v_2 \sin \omega t$ is now connected across the terminals A and B as shown in the figure.



List–I gives the possible values of ω and List–II gives different magnitudes of some quantity.

	List – I		List – II
(I)	$\omega \rightarrow 0$	(P)	0
(II)	$\omega = \frac{R}{L} = \frac{1}{RC}$	(Q)	$\frac{1}{2}$
(III)	$\omega = \frac{2R}{L} = \frac{1}{RC}$	(R)	$\frac{1}{\sqrt{2}}$
(IV)	$\omega = \frac{R}{L} = \frac{1}{\sqrt{3}RC}$	(S)	1
		(T)	$\frac{3}{4}$
		(U)	$\frac{7}{12}$

11. If $v_1 = v_2 = v_0$, then maximum values of current through resistance R (in terms of $\frac{v_0}{R}$) for different values of ω given in List-I will be

Options

(A) $I \rightarrow P, II \rightarrow Q, III \rightarrow S, IV \rightarrow R$
(B) $I \rightarrow P, II \rightarrow Q, III \rightarrow R, IV \rightarrow S$
(C) $I \rightarrow P, II \rightarrow S, III \rightarrow S, IV \rightarrow R$
(D) $I \rightarrow P, II \rightarrow S, III \rightarrow Q, IV \rightarrow T$

12. If $v_1 = v_2 = v_0$, and $\phi' = n\pi$ is the phase difference between $v_1 \cos \omega t$ and the current through the resistance then find the correct matches of n from List-II for the values of ω given in List-I.

Options

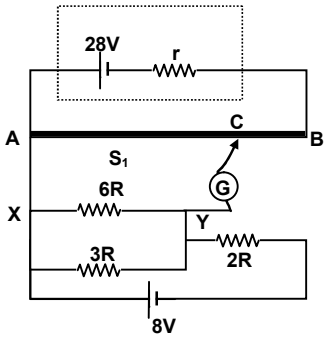
(A) $I \rightarrow T, II \rightarrow Q, III \rightarrow Q, IV \rightarrow U$
(B) $I \rightarrow P, II \rightarrow Q, III \rightarrow R, IV \rightarrow S$
(C) $I \rightarrow T, II \rightarrow S, III \rightarrow S, IV \rightarrow U$
(D) $I \rightarrow P, II \rightarrow S, III \rightarrow Q, IV \rightarrow T$

SECTION – C
(Numerical Answer Type)

This section contains **06** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

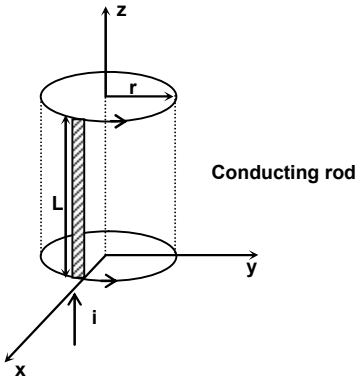
13. A gas bubble produced from an under water explosion oscillates with period T. The period T is found to be proportional to the ath power of pressure (P), b^{th} power of density (d) and c^{th} power of the total energy (E) of explosion. Using dimensional analysis find the value of (a + 4b + c).

14. In the given potentiometer circuit, the resistance per unit length (in Ω/m) 'λ' of the potentiometer wire varies with distance x (from end A) as $\lambda = 1 + 2x$. The length of the potentiometer wire is 1 m. While measuring the potential difference between the points X and Y, the balancing length (AC) is obtained exactly at the middle of the wire. The internal resistance (r) of the driving cell (in ohm) is

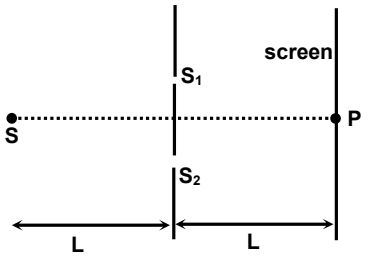


15. A ball is thrown with a speed u = $10\sqrt{3}$ m/s so that the area enclosed by the balls trajectory with the horizontal surface is maximum. Taking g = 10 m/s^2 and $\sqrt{3} = 1.732$, the maximum area (in m^2) enclosed by the balls trajectory with the horizontal surface is closest to the whole number……

16. A conducting rod of length L = $\frac{1}{4}$m is carrying a current I = 1A as shown in the diagram. It is made to move in a circular path of radius $r = \frac{2}{\pi}$m in a magnetic field $\vec{B} = B_0\hat{a}_r$ with 10 revolutions per minute (Where B_0 = 3T and $\hat{a}_r$ is a unit vector in radial direction). Find the power (in watt) required to make this rod move with a constant angular speed.



17. Light from a monochromatic point source S is incident on the two slits of an opaque wall placed at a distance L as shown in the diagram. Intensity of interference pattern is observed at the point P on the screen placed at a distance L behind the slits. When the upper slit is shifted upward then the distance between the slits increases. It is observed that when the distance between the slits is changed from 200 μm to 400 μm, the intensity at P changes from I_{max} to $\frac{3}{4}I_{max}$. If L = 1 m, then find the wave length of the light used (in Å)



18. A sound source of frequency f_0 = 1560 Hz is dropped from a height H = 505 m above the ground at t = 0. At the same time a detector is thrown upwards with a velocity u = 50 ms^{-1} along the same line. If the velocity of sound is v = 300 ms^{-1}, find the frequency (in Hz) detected by the detector at t = 5s.

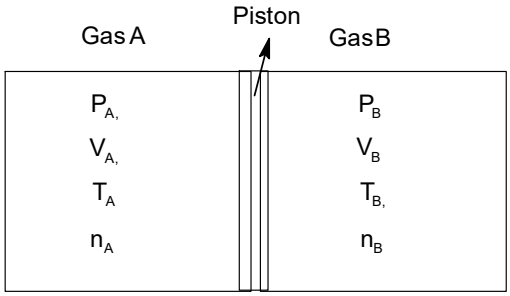
Chemistry

PART – II

SECTION – A
(One or More than one correct type)

This section contains **08** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

19. Two gas samples A and B are placed in a chamber and separated by non conducting frictionless piston.

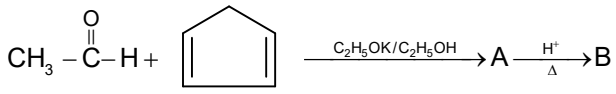


If $n_A : n_B :: 1 : 3$, $V_A : V_B :: 3 : 1$ and piston is in equilibrium then the correct statements are:
(A) $P_A = P_B$
(B) $T_A > T_B$
(C) Heat will flow from chamber 1 to chamber 2.
(D) $T_A = T_B$

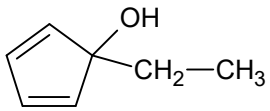
20. Borax bead test is given by the metals
(A) Cu
(B) Fe
(C) Mn
(D) Ni

21. Identify the correct statements:
(A) Lead is placed in group 1st and 2nd because it's chloride is partially soluble in water.
(B) Ferric ions give a deep green precipitate on adding $K_4[Fe(CN)_6]$ soln.
(C) As_2S_3 is soluble in yellow ammonium sulphide solution.
(D) Heating of NH_4Cl with NaOH form N_2 gas.

22. Compounds having white colour.
(i) ZnS (ii) CdS (iii) $PbCl_2$ (iv) SnS_2 (v) $AlOH)_3$
(vi) $BaCO_3$ (vii) $SrCO_3$ (viii) $Mg(NH_4)HPO_4$
(A) $Al(OH)_3$, SnS_2, $BaCO_3$
(B) $BaCO_3$, $Mg(NH_4)HPO_4$
(C) $SrCO_3$, $CaCO_3$
(D) ZnS, $PbCl_2$, $BaCO_3$

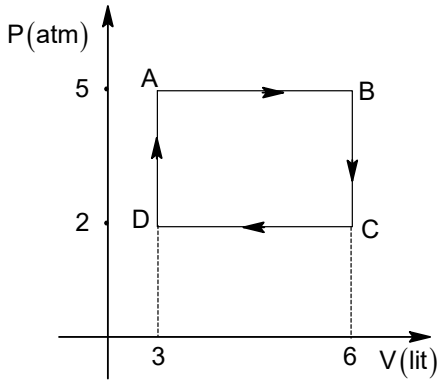
23. 

Find the correct options for the above

(A) Compound A will be 

(B) Compound B will be toluene
(C) Compound B give etard's reaction
(D) Compound A is not an aromatic compounds.

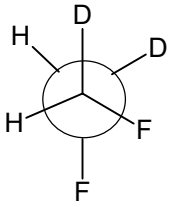
24. For 2 mole of He gas (assuming ideal behaviour), study P – V curve.

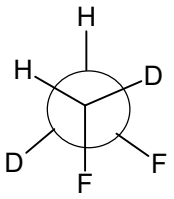


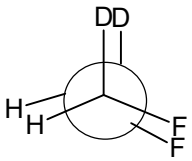
For the given system
(A) Process AB, $\Delta U > 0$
(B) For process BC, $\Delta T > 0$
(C) For process CD, $\Delta Q < 0$
(D) For process DA, $\Delta Q > 0$

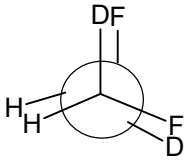
25. Identify the correct statements related to alloy with their uses?
(A) Brass is used in the cartridges cases.
(B) Bronze is used in the formation of bells.
(C) Phosphorous bronze is used in propellers.
(D) Gun metal is used in preparation of T.V.

26. Identify the identical compounds?

(A)


(B)


(C)


(D)


(Matching List Type)

This section contains **TWO (02)** List-Match Sets. Each List-Match set has **TWO(02)** Multiple Choice Questions. Each List-Match set has two lists: **List-I** and **List-II**. **List-I** has **Four** entries (I), (II), (III) and (IV) and **List-II** has **Six** entries (P), (Q), (R), (S), (T) and (U). **FOUR** options are given in each Multiple Choice Question based On **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.

Answer the following by appropriately matching the list based on the information given in the paragraph.

Every polymers has some uses, so based on the uses (List –II) of polymers (List –I) given here.

List -I (Polymer)		List II (Uses)	
(I)	Glyptal	(P)	Radio cabinets
(II)	Bakelite	(Q)	Laminated sheets
(III)	Polystyrene	(R)	Fibres
(IV)	Urea – formaldehyde resin	(S)	Unbreakable cups
		(T)	Lacquers
		(U)	Computer discs

27. Which of the following options have correct combinations considering List-I and List-II
(A) I – (S)
(B) III – (Q)
(C) II – (U)
(D) IV – (R)

Every polymers has some uses, so based on the uses (List –II) of polymers (List –I) given here.

List -I (Polymer)		List II (Uses)	
(I)	Glyptal	(P)	Radio cabinets
(II)	Bakelite	(Q)	Laminated sheets
(III)	Polystyrene	(R)	Fibres
(IV)	Urea – formaldehyde resin	(S)	Unbreakable cups
		(T)	Lacquers
		(U)	Computer discs

28. Which of the following options has the correct combination considering List-I and List-II?
(A) IV – (U)
(B) III – (P)
(C) I – (S)
(D) II – (Q)

Answer the following by appropriately matching the list based on the information given in the paragraph.

Ore contain metals with some other elements. List –I contains ores and List –II contain element present in that

List -I		List II	
(I)	Dolomite	(P)	C
(II)	Limonite	(Q)	Cu
(III)	Azurite	(R)	Al
(IV)	Alunite	(S)	S
		(T)	Fe
		(U)	Mg

29. Which of the following options has correct combination considering List –I and List –II?
(A) I – (T)
(B) IV – (T)
(C) III – (Q)
(D) II – (R)

Ore contain metals with some other elements. List –I contains ores and List –II contain element present in that

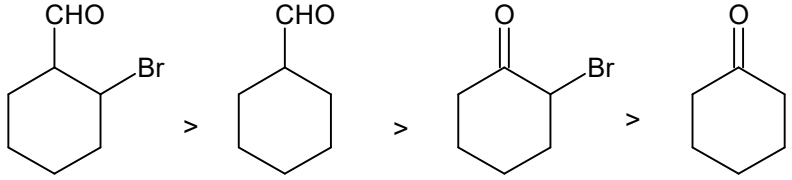
List -I		List II	
(I)	Dolomite	(P)	C
(II)	Limonite	(Q)	Cu
(III)	Azurite	(R)	Al
(IV)	Alunite	(S)	S
		(T)	Fe
		(U)	Mg

30. Which of the following options has correct combinations considering List –I and List –II?
(A) IV – (P)
(B) II – (T)
(C) III – (R)
(D) IV – (Q)

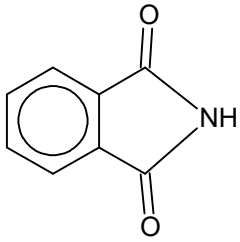
SECTION – C
(Numerical Answer Type)

This section contains **06** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

31. 'x' are the total number of correct orders and 'y' are the total number of incorrect orders as properties mentioned. Find out x/y?

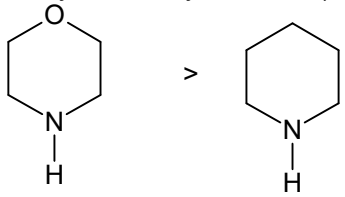
(a) 

(Extent of hydration toward nucleophilic addition reaction)

(b) $CH_3-CH_2-NH_2 < CH_3-\overset{\overset{O}{\|}}{C}-NH_2 <$ 

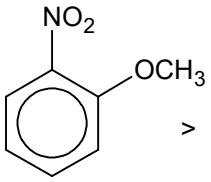
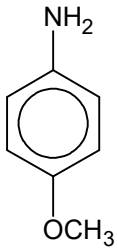
(acidic strength)

(c) Pyridine > Pyrrole (Basic strength)

(d) 

(Basic strength)

(e) N-methyl aniline > Aniline (Basic strength)

(f)  >  (Basic strength)

(g) $PhC \equiv CH_2$ and $PhCH = CH_2$ can be differentiated by Br_2/CCl_4 test.

(h) 1-Butyne and 2-butyne can be distinguished by ammonical $AgNO_3$ solution.

(i) Compound containing $-\overset{OH}{\underset{|}{C}}H-\overset{O}{\overset{||}{C}}-$ group and that containing $-\overset{O}{\overset{||}{C}}-H$ group can be distinguished by Tollen's reagent.

32. Carbon crystalizes into its allotrope diamond. If x is percentage of area occupied by carbon atoms in one face of a unit cell then what is value of $\frac{2x}{\pi}$?

33. $\underset{C_4H_8O}{A} \xrightarrow{HI(excess)} B \xrightarrow{alc.\ KOH} C \xrightarrow[(ii)\ H_2O+Zn]{(i)\ O_3} D + E$

Compound A is cyclic optically active ether. Both D and E compounds give Cannizzaro reaction. Number of carbon in compound E is greater than D. What is ratio of total number of atoms in compound E with total number of atoms in compound D.

34. Total number of maximum possible visible lines when electrons jump from energy level n = 10 to n = 1 for sample of hydrogen atoms are 'y'. What is the value of y/5?

35. Ratio of number of incorrect reactions with the number of correct reactions is:

(A) $CsI_3 \xrightarrow{heat} CsI + I_2$

(B) $AlCl_3 + \underset{(melt)}{ICl} \longrightarrow I^+ + [AlCl_4]^-$

(C) $UF_4 + ClF_3 \longrightarrow UF_6 + ClF$

(D) $XeF_4 + 2SF_4 \longrightarrow Xe + 2SF_6$

(E) $4KClO_3 \xrightarrow{low\ T} 3KClO_4 + KCl$

36. Total number of compounds which contain bond angle (E – O – E) greater than $109^o\ 28'$ are 'x'.
Cl_2O, CH_3OCH_3, H_2O, $Cr_2O_7^{-2}$, Br_2O, C_2H_5OH, C_6H_5OH, CH_3OH, I_3^-.
Find out the x/2?

Mathematics

PART – III

SECTION – A
(One or More than one correct type)

This section contains **08** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

37. In a ΔABC, $A \equiv (\alpha,\beta)$, $B \equiv (1,2)$, $C \equiv (2,3)$ and point A lies on $y = 2x + 3$, where $\alpha, \beta \in I$ and area S of the triangle is such that $2 \le S < 3$. Then coordinates of A can be
(A) $(-6,-9)$
(B) $(-7,-11)$
(C) $(2,7)$
(D) $(3,9)$

38. From the intersection point of the circle $x^2 + y^2 = 9$ and the line $y = 2x$ perpendicular is drawn to the major axis of the ellipse $\frac{x^2}{9} + \frac{y^2}{4} = 1$, which meets the ellipse at point M, then equation of tangent at M to the ellipse is/are
(A) $x + 3y - 3\sqrt{5} = 0$
(B) $4x + 3y - \sqrt{5} = 0$
(C) $x + 3y + 3\sqrt{5} = 0$
(D) $4x + 3y + \sqrt{5} = 0$

39. The equation $\left|\sqrt{(x-4)^2 + (y-2)^2} - \sqrt{(x+4)^2 + (y+2)^2}\right| = 4$ represents
(A) Ellipse with centre (0, 0)
(B) Hyperbola with centre (0, 0)
(C) Conic with eccentricity $\sqrt{5}$
(D) Conic whose auxillary circle is $x^2 + y^2 = 4$

40. If PQ is a chord of parabola $x^2 = 4y$ which subtends right angle at the vertex. Then locus of centroid of triangle PSQ (S is focus) is a parabola whose
(A) vertex is $(0,3)$
(B) Length of latus rectum is $\frac{4}{3}$
(C) axis is $x = 0$
(D) Tangent at the vertex is $x = 3$

41. If $(1 + x + x^2 + x^3)^{100} = a_0 + a_1x + a_2x^2 + ... + a_{300}x^{300}$, then
(A) $a_0 + a_1 + a_2 + a_3 + ... + a_{300}$ is divisible by 1024
(B) $a_0 + a_2 + a_4 + ... + a_{300} = a_1 + a_3 + ... + a_{299}$
(C) $a_r - a_{300-r} = 0$, $r = \{0, 1, 2, ..., 300\}$
(D) $a_1 = 100$

42. If z is the complex number satisfying $|z-3-4i| \le 10$ and $\alpha = \sin^{-1}\left(\sin|z|_{max}\right)$, $\beta = \cos^{-1}\left(\cos\left(-\frac{|z|_{max}}{3}\right)\right)$, then

(A) $\alpha - 3\beta = \pi$

(B) $\alpha - 3\beta = -\pi$

(C) $\sin^2\alpha + \cos^2 3\beta = 1$

(D) $\sin^2\alpha + \cos^2 3\beta = -1$

43. The correct statement(s) is/are

(A) The lines of intersection of planes $\vec{r}.\vec{n}_1 = q_1$, $\vec{r}.\vec{n}_2 = q_2$ and $\vec{r}.\vec{n}_3 = q_3$, $\vec{r}.\vec{n}_4 = q_4$ are perpendicular if $(\vec{n}_1.\vec{n}_3)(\vec{n}_2.\vec{n}_4) = (\vec{n}_1.\vec{n}_4)(\vec{n}_2.\vec{n}_3)$.

(B) If three distinct planes $\vec{r}.\vec{n}_1 = q_1$, $\vec{r}.\vec{n}_2 = q_2$, $\vec{r}.\vec{n}_3 = q_3$ intersect in a line which is contained by the plane $\vec{r}.\vec{n}_4 = q_4$, then $[\vec{n}_1 \quad \vec{n}_2 \quad \vec{n}_4]\vec{n}_3 = [\vec{n}_1 \quad \vec{n}_2 \quad \vec{n}_3]\vec{n}_4$

(C) If four distinct planes $\vec{r}.\vec{n}_1 = q_1$, $\vec{r}.\vec{n}_2 = q_2$, $\vec{r}.\vec{n}_3 = q_3$, $\vec{r}.\vec{n}_4 = q_4$ intersect in a line then, $[\vec{n}_1 \quad \vec{n}_2 \quad \vec{n}_4]\vec{n}_3 = [\vec{n}_1 \quad \vec{n}_2 \quad \vec{n}_3]\vec{n}_4$

(D) If a plane contains line of intersection of planes $\vec{r}.\vec{n}_1 = q_1$, $\vec{r}.\vec{n}_2 = q_2$ and is parallel to the line of intersection of the planes $\vec{r}.\vec{n}_3 = q_3$, $\vec{r}.\vec{n}_4 = q_4$, then $[\vec{n}_1 \quad \vec{n}_2 \quad \vec{n}_4]\vec{n}_3 = [\vec{n}_1 \quad \vec{n}_2 \quad \vec{n}_3]\vec{n}_4$

44. Points that lie on the lines bisecting the angles between the lines $\frac{x-2}{2} = \frac{y-3}{3} = \frac{z-6}{6}$ and $\frac{x-2}{3} = \frac{y-3}{6} = \frac{z-6}{2}$ are

(A) $(7,12,14)$

(B) $(0,-3,14)$

(C) $(1,0,10)$

(D) $(-3,-6,-2)$

(Matching List Type)

This section contains **TWO (02)** List-Match Sets. Each List-Match set has **TWO(02)** Multiple Choice Questions. Each List-Match set has two lists: **List-I** and **List-II**. **List-I** has **Four** entries (I), (II), (III) and (IV) and **List-II** has **Six** entries (P), (Q), (R), (S), (T) and (U). **FOUR** options are given in each Multiple Choice Question based On **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.

Answer the following by appropriately matching the list based on the information given in the paragraph.

MM-I. Match the following list(s)

LIST-I		**LIST-II**	
(I)	Number of integers in the range of the function $f(x) = \cos 2x - 10\cos x - 11$ is	(P)	10
(II)	The period of the function $f(x) = \left(\sec^2\left(\frac{\pi x}{10}\right) - \tan^2\left(\frac{\pi x}{10}\right)\right)^{\cos^4 4\pi x + 100x - [100x]}$ (where [.] denotes greatest integer function) is	(Q)	14
(III)	If the number of the solution of the equation $\sin^{-1}(\sin x) - mx = 0$, $(m > 0)$ is 5, then 70m is equal to	(R)	21
(IV)	The period of the function $f(x) = \sum_{r=1}^{n}\left[r^2 + e^{-x^2-1} - r + 1\right]$ is equal to (where [.] denotes greatest integer function)	(S)	28
		(T)	29
		(U)	37

45. Which of the following is the only CORRECT combination?
(A) (I), (P)
(B) (II), (Q)
(C) (III), (R)
(D) (IV), (S)

46. Which of the following is the only INCORRECT combination?
(A) (I), (R)
(B) (II), (S)
(C) (IV), (P)
(D) (IV), (T)

Answer the following by appropriately matching the list based on the information given in the paragraph.

MM-II. Match the following list(s)

LIST-I		LIST-II	
(I)	The number of point(s) (x,y), (where both x and y are perfect squares of integers) on the parabola $y^2 = px$, p being a prime number, is	(P)	1
(II)	If AFB is a focal chord of the parabola $y^2 = 4ax$ and AF = 2, FB = 6 then the length of latus rectum is (where F is focus of the parabola)	(Q)	4
(III)	The coordinates of the point on the parabola $y = x^2 + 7x + 2$ which is nearest to the straight line $y = 3x - 3$ is (a,b) the $2a - b$ is equal to	(R)	6
(IV)	The value of a for which at least one tangent to the parabola $y^2 = 4ax$ becomes normal to the circle $x^2 + y^2 - 2ax - 4ay + 3a^2 = 0$ is	(S)	8
		(T)	9
		(U)	12

47. Which of the following is the only CORRECT combination?
(A) (I), (Q)
(B) (II), (P)
(C) (III), (S)
(D) (IV), (R)

48. Which of the following is the only INCORRECT combination?
(A) (I), (P)
(B) (II), (R)
(C) (III), (S)
(D) (IV), (T)

SECTION – C
(Numerical Answer Type)

This section contains **06** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

49. Let $\vec{x} = \hat{i} + 3\hat{j} - 2\hat{k}$, $\vec{y} = 2\hat{i} + 4\hat{j} + \hat{k}$ and a vector $\vec{z}$ satisfying $\vec{x} \times \vec{z} = \vec{x} \times \vec{y}$ and $\vec{z}.\vec{x} = 0$, then $\left[|\vec{z}|_{max}\right]$ is equal to (where [.] denotes greatest integer function)

50. Let $2[x]$, $6\{x\}$ and $3x$ be first three terms of a G.P. where $x \in R^+$ and (where [.] and {.} represents greatest integer and fractional part functions respectively). If there is another G.P. whose terms are square of reciprocal of the terms of the given G.P. and sum of infinite terms of this G.P. is S, then 100S is equal to

51. If a, b, c, d are prime numbers such that $a^2 - b^2 = c$ and $d = b + c$, then sum of all the 4 digit numbers formed by using the digits a, b, c, d without repetition is λ, then $\frac{\lambda}{10}$ is

52. Number of solution(s) of the equation $\cos 2x = \cos^3 x - \sin^3 x$ in $[0, 2\pi]$ is/are

53. If $x = 2\tan 10^\circ + \tan 50^\circ$, $y = \tan 20^\circ + \tan 50^\circ$, $w = 2\tan 10^\circ + \tan 70^\circ$, $z = \tan 20^\circ + \tan 70^\circ$, then the value of $\left(\frac{z - 2y}{x + w}\right)$ is equal to

54. If z is a complex number satisfying $(2 + 3i)z + (2 - 3i)\bar{z} - 6 = 0$, then minimum value of $[13|z|]$ is (where [.] denotes greatest integer function)